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Verbal Ability

Instructions [1 - 2]

In each of the questions a word has been used in sentences in four different ways. Choose the option corresponding to the sentence in which the usage of the word is incorrect or inappropriate.

1. Complement

- A** The airline provided guests with gift hampers as a complement to their loyalty.
- B** The new curtains that were used were a complement to the walls.
- C** The economic crisis of that decade had its complement in a political crisis in the next.
- D** The objective of that session was to teach which colour complement each other.

Answer: C

2. Glow

- A** I saw her glow with pride as she described her son's achievements.
- B** The glow on her face was testimony to her embarrassment.
- C** The bonfire cast a glow on the surrounding darkness.
- D** The glow on her face was an indication of her sullenness.

Answer: B

Instructions [3 - 5]

Arrange the sentences in the most logical order to form a coherent paragraph. From the given options (A, B, C, D) choose the appropriate sequence.

- 3.
 - i. Big Tobacco is doing nothing illegal by producing and marketing cigarettes.**
 - ii. Even so, regulators weighing up how to treat safer alternatives to cigarettes are often too harsh.**
 - iii. No wonder people are cynical when they hear tobacco bosses evangelise about e-cigarettes.**
 - iv. The industry has an inglorious history of lying about the effects of cigarettes on human health.**
- A** (i), (iv), (iii), (ii)
 - B** (iv), (iii), (ii), (i)
 - C** (i), (ii), (iv), (iii)
 - D** (iv), (i), (iii), (ii)

Answer: A

4. i. The data came from the UK Biobank, which contains genetic and medical data from half a million people.
ii. Intriguingly, this analysis suggests genetic contributions to intelligence and educational achievement are currently disfavoured by natural selection.
iii. Positive correlation means an association with successful reproduction; negative one means exactly the opposite.
iv. A study just published uses a new statistical method to examine how genetic contributions to certain human traits such as intelligence correlate With how many children a person has.

- A** (iii), (ii), (iv), (i)
B (iv), (i), (iii), (ii)
C (iv), (ii), (iii), (i)
D (iv), (i), (ii), (iii)

Answer: D

5. i. I am confident that you gentlemen will review without passion the evidence you have heard come to a decision, and restore this defendant to his family.
ii. I'm no idealist to believe firmly in the integrity of our courts and in the jury system that iS no ideal to me, it is a living, working reality.
iii. A court is only as sound as its jury, and a jury is only as sound as the men who make it up.
iv. Gentlemen, a court is no better than each man of you sitting before me on this jury.

- A** (ii), (iv), (iii), (i)
B (i), (ii), (iii), (iv)
C (iv), (iii), (i), (ii)
D (iii), (iv), (i), (ii)

Answer: A

Instructions [6 - 9]

Read the passage and choose the most appropriate answer for the question that follow.

Passage:

In my research on leadership transitions, I have observed that career advances require all of us to move way beyond our comfort zones. At the same time, however, they trigger a strong countervailing impulse to protect our identities: When we are unsure of ourselves or our ability to perform well or measure up in a new setting, we often retreat to familiar behaviors and styles.

But my research also demonstrates that the moments that most challenge our sense of self are the ones that can teach us the most about leading effectively. By viewing ourselves as works in progress and evolving our professional identities through trial and error, we can develop a personal style that feels right to us and suits our organizations' changing needs.

That takes courage, because learning, by definition, starts with unnatural and often superficial behaviors that can make us feel calculating instead of genuine and spontaneous. But the only way to avoid being

pigeonholed and ultimately become better leaders is to do the things that a rigidly authentic sense of self would keep us from doing.

The word “authentic” traditionally referred to any work of art that is an original, not a copy. When used to describe leadership, of course, it has other meanings—and they can be problematic. For example, the notion of adhering to one “true self” flies in the face of much research on how people evolve with experience, discovering facets of themselves they would never have unearthed through introspection alone. And being utterly transparent—disclosing every single thought and feeling—is both unrealistic and risky.

6. **From the passage, we can infer that**

- A** the author believes that in order to be authentic one should remain true to oneself
- B** the author believes that authentic could prevent us from trying out new styles
- C** authenticity is an enabling characteristic in the context of leadership
- D** authenticity involves spontaneous behaviour that leads to risky outcomes

Answer: B

7. **The term "work in progress" in the passage, refers to**

- A** an assumption that individual have a lot of growing up to do
- B** a belief that good leaders do not have areas where they can improve
- C** an assumption that all humans progress and change for the better
- D** a belief that individual can change as they encounter new experiences

Answer: D

8. **For the author, a leadership transition requires a manager to**

- A** change and adapt to new role requirements
- B** avoid being stereotyped by others
- C** not share all his thoughts and feelings
- D** avoid copying other's styles

Answer: A

9. **The most appropriate title for this passage would be**

- A** Leadership Planning
- B** The Authenticity Paradox
- C** Can One be True to Oneself?

D Works in Progress

Answer: A

Instructions [10 - 15]

Read the passage and choose appropriate answer for the questions that follow.

Passage:

As rough sleeping rises nationally, the exact scale of the crisis remains hard to capture. The official data shows that in England, rough sleeping has risen for six years in a row. The latest figures estimated that 4,134 people bedded down outside in 2016, up 16% on the previous year. Though London remains the centre of rough sleeping, accounting for 23% of the national total (and in Westminster, with 260 rough sleepers, the highest number of cases), the rate is increasing much faster outside the capital, in places such as Brighton, Manchester and Birmingham.

Each winter across the country, councils send teams of volunteers to conduct night-time counts of all the rough sleepers in the borough to assess how acute the problem is. Recent counts in the homelessness hotspots of Cambridge and Hackney, east London, reveal how the problem is evolving.

"This is their bedroom as you are entering. Be respectful of that," warned the organiser of the Cambridge count, before teams set out to count rough sleepers in the historic centre in early Friday. For bedroom, read shop doorway, church graveyard, or multi-storey car park - anywhere in the cold night air where a street sleeper might hope to find a yard or two of dry shelter and, if they are lucky, a degree of privacy.

At 3 am, as the last of the evening's city-centre revellers are going home, the teams set out. This is the time when rough sleepers consider it safe enough and sufficiently quiet to bed down. Dotted along a line of shops on a main shopping street were several people in brightly coloured sleeping bags in doorways, surrounded by the paraphernalia of street life: plastic bags stuffed with belongings, cardboard under sheets to insulate them from the cold, the odd half-empty wine bottle.

There are strict definitions of what constitutes a rough sleeper for the purposes people must be sleeping, about to bed down or bedded down on the ground: doorways, parks, tents, bus shelters, cars, barns, sheds and other street-side spaces not designed for habitation. Homeless people who are resident in hostels or shelters on places the night in question are not counted. The count is not a precise science: bad weather can depress the figures; counters can miss rough sleepers if they are well hidden; regular sleepers may by chance spend the night elsewhere. Good housing support services, too, can have a positive effect in reducing the numbers.

As the main city within a large rural area, and one with good homelessness provision such as hostels, Cambridge has always acted as a magnet for rough sleepers. Relationship breakdown and substance abuse remain important triggers of homelessness. But increasingly other factors have come into play, not least poverty: the lack of affordable housing, high rents and unstable tenancies, housing benefit cuts, and precarious incomes caused by the rise of zero-hours working.

Exactly how bad the problem has got is a matter for debate. Between October and the end of November each year, every English local authority is required to submit snapshot estimates of the number of people sleeping out on a specified night.

10. The term 'rough sleeping', as used in this passage refers to

A people roughing it out camping outdoors

B people sleeping in homeless shelters

- C** people sleeping in public spaces not earmarked for the purpose
- D** homeless people sleeping uncomfortably after roughing others up

Answer: C

11. **According to the author, all of the following are causes of rough sleeping except:**

- A** poor housing support services
- B** revelling in the city centre in the evening
- C** relationship breakdown and substance abuse
- D** high rents and unstable tenancies

Answer: B

12. **The scale of rough sleeping is difficult to capture because**

- A** the rate is increasing much faster outside London
- B** rough sleepers insist on a degree of privacy
- C** there are strict definitions of what constitutes a rough sleeper
- D** weather conditions or hidden locations can skew the figures

Answer: D

13. **This article primarily focuses on rough sleeping in the city of**

- A** London
- B** Westminster
- C** Cambridge
- D** Birmingham

Answer: C

14. **The most appropriate title for this article is**

- A** Counting Rough-sleepers in England's Homeless Hotspots
- B** Be Respectful of Peoples' Bedrooms in England's Homeless Hotspots
- C** The Scale of Rough Sleeping: Reasons and Solutions
- D** An Analysis of Rough Sleeping: Towards a Solution

Answer: A

15. The organisers' warning to be respectful refers to

- A** the services provided by parks, bus shelters, barns and sheds for the rough sleepers
- B** the need for a degree of privacy for the rough sleepers
- C** concern for relationship breakdowns and substance abuse among the rough sleepers
- D** Deference for the brightly coloured sleeping bags owned by rough sleepers

Answer: B

Logical Reasoning & Data Interpretation

16. A club with x members is organised into four committees according to the following rules:

- (i) Each member belongs to exactly two committees.
- (ii) Each part of committees has exactly one member in common. Then

- A** $x = 4$
- B** $x = 6$
- C** $x = 8$
- D** x cannot be determined

Answer: B

17. Let P , Q , R and S be statements such that

- (i) if both P and Q are true then R is false and
- (ii) if P is false then S is false

Suppose R is true. Then which of the following necessarily holds?

- A** Q is false
- B** if S is false then Q is false
- C** if S is true then Q is true
- D** if S is true then Q is false

Answer: D

Explanation:

Since R is true, then, both P and Q cannot be true together.

Also, using statement (ii), we can draw out the following cases:

P	Q	R	S
False	True	False	False
True	False	False	True
True	False	False	False
False	False	False	False

Now, evaluating the given options,

only option 4 is true in all the four cases.

Instructions [18 - 20]

Answer the question based on the following information.

Houses numbered 1 to 4 are situated east to west in that order. The houses are each occupied by professorsofa college, namely, Prof.Sinha, Prof.Khanduja, Prof. Saxena and Prof.Kesarwani. They teach different languages: Urdu, Bengali, Gujarati and Sanskrit, and possess different makes of motorcycles each manufactured in a different year. Each of the professors teaches only one subject and owns only one motorcycle. The following additional information is available:

- * The owner of the Suzuki motorcycle teaches Gujarati
- * Prof. Anshul Sinha has a Honda motorcycle and teaches Urdu
- * Prof.Khanduja, who teaches Gujarati, lives in house number 2
- * Prof. Saxena teaches Sanskrit and lives in the western-most house
- * 2001 model of motorcycle is owned by the professor of Urdu who lives in house number 1
- * Prof.Kesarwani owns a BMW motorcycle

18. Which of the following statements is incorrect?

- A** Prof.Kesarwani lives in house number 1
- B** Prof.Sinha owns a 2001 model of motorcycle
- C** Prof.Khaduja rides a Suzuki motorcycle
- D** Prof.Khaduja has two neighbours

Answer: A

Explanation:

Subject	Bike	Professor	House No.
Urdu	Honda	Anshul	01
Bengali	BMW	Kesarwani	03
Gujrati	Suzuki	Khanduja	02
Sanskrit		Saxena	04

Prof. Kesarwani lives in house number 3. So, statement 1 is false.

19. From the above information we can infer that

- A** Prof.Saxena and Khanduja are neighbours

- B** Prof.Kesarwani teaches Bengali
- C** Prof.Khaduja drives a 2001 model of motorcycle
- D** Prof.Sinha rides a Suzuki motorcycle

Answer: B

Explanation:

We will proceed by creating a table:

The western-most house is numbered as 4 as Houses numbered 1 to 4 are situated east to west in that order.

Subject	Bike	Professor	House No.
Urdu	Honda	Anshul	01
Bengali	BMW	Kesarwani	03
Gujrati	Suzuki	Khanduja	02
Sanskrit		Saxena	04

Option 2 is true.

20. **Prof.Saxens drives motorcycle of the following make**

- A** Honda
- B** BMW
- C** Suzuki
- D** none of the above

Answer: D

Explanation:

We will use tabular form to solve this question:

Subject	Bike	Professor	House No.
Urdu	Honda	Anshul	01
Bengali	BMW	Kesarwani	03
Gujrati	Suzuki	Khanduja	02
Sanskrit		Saxena	04

No information is given about Mr. Saxena's bike. So, None of these.

Instructions [21 - 25]

Answer the question based on the following information.

Table 1 gives the men's and women's world record times for various outdoor running distances, recognized by the International Association of Athletics Federations (IAAF) as of 17 November 2017.

Table 1: World record running times data

Running distance (D) (meters (m))	Men's record (M) (seconds (s))	Women's record (W) (seconds (s))
100	9.6	10.5
200	19.2	21.3
400	43	47.6
800	101	113.3
1000	132	149
1500	206	230.1
2000	284.8	323.8
3000	440.7	486.1
5000	757.4	851.2
10000	1577.5	1757.5

21. The average speeds (rounded to 2 places of decimal) for the men's and women's fastest 1500 meters race have been, respectively.

- A 7.28 m/s and 6.52 m/s
- B 7.32 m/s and 6.52 m/s
- C 7.28 m/s and 6.48 m/s
- D 7.32 m/s and 6.48 m/s

Answer: A

Explanation:

For Men's 1500 m, fastest time recorded = 206 seconds.

So, average speed = $\frac{\text{Total distance}}{\text{Total time}} = \frac{1500}{206} = 7.28 \text{ metres per sec.}$

For Women's 1500 m, fastest time recorded = 230.1

So, average speed = $\frac{\text{Total distance}}{\text{Total time}} = \frac{1500}{230.1} = 6.52 \text{ metres per sec.}$

Therefore, Option A.

22. For which of the following categories, is the women's world record time less than 110% of the men's world record time?

- A 800 meters
- B 400 meters
- C 200 meters
- D 100 meters

Answer: D

Explanation:

Using the given options, we create the following table:

Running Distance (D)	Men's Record (t)	110% of t	Women's Record (t')
100	9.6	10.56	10.5
200	19.2	21.12	21.3
400	43	47.3	47.6
800	101	111.1	113.3

We can observe that only in case of 100 meters is Women's record time is less than 110% of the men's record time.

Hence, Option D.

23. For which of the following categories is the ratio of the women's and men's record times, that is, W/M, the largest?

- A 100 meters
- B 200 meters
- C 800 meters
- D 2000 meters

Answer: D

Explanation:

Using the table as shown below:

Running Distance (D)	Men's Record (t)	Women's Record (t')	W/M
100	9.6	10.5	1.09
200	19.2	21.3	1.11
400	43	47.6	1.11
800	101	113.3	1.12
1000	132	149	1.13
1500	206	230.1	1.12
2000	284.8	323.8	1.14
3000	440.7	486.1	1.10
5000	757.4	851.2	1.12
10000	1577.5	1757.5	1.11

We can infer that the ratio W/M is greatest for 2000 metres.

24. The category in which the average speed is second highest is

- A Men's 100 meters
- B Men's 200 meters
- C Women's 100 meters
- D Women's 200 meters

Answer: C

Explanation:

Using the formula Speed= Distance/ Time we can formulate the following table:

Running Distance (D)	Men's Record (t)	Men's speed (D/t)	Women's Record (t')	Women's Speed (D/t')
100	9.6	10.42	10.5	9.52
200	19.2	10.42	21.3	9.39
400	43	9.30	47.6	8.40
800	101	7.92	113.3	7.06
1000	132	7.58	149	6.71
1500	206	7.28	230.1	6.52
2000	284.8	7.02	323.8	6.18
3000	440.7	6.81	486.1	6.17
5000	575.4	8.69	851.2	5.87
10000	1577.5	6.34	1757.5	5.69

From the above table, we can infer that the highest speed is that in case of men 100 and 200 metres race.

The second highest speed is in case of women 100 metre race.

25. If α is the factor by which the women's world record average speed reduces with doubling of the running distance, the smallest value of α occurs for the pair of categories

- A (100m, 200m)
- B (1500 m, 3000m)
- C (5000 m, 10000m)
- D (400 m, 800m)

Answer: A

Explanation:

Let us evaluate each of the options one by one.

In the first option, we need to check the speeds for 100 m and 200 m.

For 100 m, Average speed in women's category = $\frac{\text{Total distance}}{\text{Total time}} = \frac{100}{10.5} = 9.52 \text{ metres per sec.}$

For 200 m, Average speed in women's category = $\frac{\text{Total distance}}{\text{Total time}} = \frac{200}{486.1} = 9.39 \text{ metres per sec.}$

$$\alpha = 1 - \frac{9.39}{9.52} = 1 - 0.9863 = 0.0137$$

In the second option, we need to check the speeds for 1500 m and 3000 m.

For 1500 m, Average speed in women's category = $\frac{\text{Total distance}}{\text{Total time}} = \frac{1500}{230.1} = 6.52 \text{ metres per sec.}$

For 3000 m, Average speed in women's category = $\frac{\text{Total distance}}{\text{Total time}} = \frac{3000}{586.1} = 6.17 \text{ metres per sec.}$

$$\alpha = 1 - \frac{6.17}{6.52} = 1 - 0.9463 = 0.0536$$

In the third option, we need to check the speeds for 5000 m and 10000 m.

For 5000 m, Average speed in women's category = $\frac{\text{Total distance}}{\text{Total time}} = \frac{5000}{851.2} = 5.874 \text{ metres per sec.}$

For 10000 m, Average speed in women's category = $\frac{\text{Total distance}}{\text{Total time}} = \frac{10000}{1757.5} = 5.6912 \text{ metres per sec.}$

$$\alpha = 1 - \frac{5.6912}{5.874} = 1 - 0.9688 = 0.0311$$

In the fourth option, we need to check the speeds for 400 m and 800 m.

For 400 m, Average speed in women's category = $\frac{\text{Total distance}}{\text{Total time}} = \frac{400}{47.6} = 8.40336 \text{ metres per sec.}$

For 800 m, Average speed in women's category = $\frac{\text{Total distance}}{\text{Total time}} = \frac{800}{113.3} = 7.0609 \text{ metres per sec.}$

$$\alpha = 1 - \frac{7.0609}{8.40336} = 1 - 0.84024 = 0.16$$

Therefore, lowest α is in the first option

Quantitative Aptitude

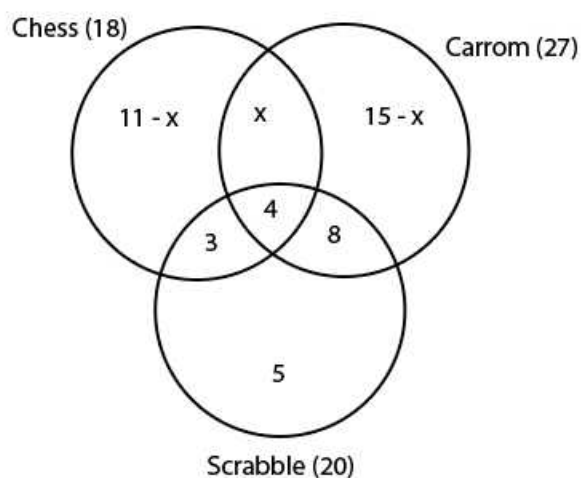
26. Each student in a class of 40 plays at least one indoor game-chess, carom and scrabble. 18 play chess, 20 play scrabble and 27 play carom. 7 play both chess and scrabble, 12 play both scrabble and carom and 4 play all 3 games. The number of players who play chess and carom but not scrabble is

- A 10
- B 4
- C 6
- D 12

Answer: C

Explanation:

Students who play all the three games are 4, and with the rest information given in the question, we can at best draw the venn diagram as:



Now, total students=40.

So, Chess+ 5+8+15-x=40

$$\Rightarrow 18+28-x=40$$

$$\Rightarrow 46-x=40$$

$$\therefore x=6$$

27. The value of $20C_1 + 2 \times 20C_2 + 3 \times 20C_3 + \dots + 20 \times 20C_{20}$ is

- A 19×20^{20}
- B 20×2^{19}

C 20×2^{20}

D 19×2^{19}

Answer: B

Explanation:

Let $S = (0 \times {}^{20}C_0) + (1 \times {}^{20}C_1) + (2 \times {}^{20}C_2) + (3 \times {}^{20}C_3) + \dots + (19 \times {}^{20}C_{19}) + (20 \times {}^{20}C_{20})$.-----Eqn 1.

When we reverse the RHS and write again, we get

$S = (20 \times {}^{20}C_{20}) + (19 \times {}^{20}C_{19}) + (18 \times {}^{20}C_{18}) + (17 \times {}^{20}C_{17}) + \dots + (1 \times {}^{20}C_1) + (0 \times {}^{20}C_0)$.-----Eqn 2.

When we use ${}^nC_r = {}^nC_{n-r}$, we can re-write the second equation as:

$S = (20 \times {}^{20}C_0) + (19 \times {}^{20}C_1) + (18 \times {}^{20}C_2) + (17 \times {}^{20}C_3) + \dots + (1 \times {}^{20}C_{19}) + (0 \times {}^{20}C_{20})$.-----Eqn 3.

Adding Eqn 1 and Eqn 3, we will get,

$$2S = 20({}^{20}C_0 + {}^{20}C_1 + {}^{20}C_2 + {}^{20}C_3 + \dots + {}^{20}C_{19} + {}^{20}C_{20})$$

$$\Rightarrow 2S = 20 \times 2^{20}$$

$$\Rightarrow S = \frac{20 \times 2^{20}}{2}$$

$$\therefore S = 20 \times 2^{19}$$

28. The circle $x^2 + y^2 = 8$ intersects the parabola $y^2 = 2x$ at a point P in the first quadrant. The acute angle between the tangents to the circle and the parabola at the point P is

A $\tan^{-1}(3)$

B $\tan^{-1}(-3)$

C $\tan^{-1}(\sqrt{3})$

D $\tan^{-1}(-\sqrt{3})$

Answer: A

Explanation:

First, we will try and find the point of intersection of the two curves.

We do so by equating the two curves.

$$\text{So, } x^2 + y^2 - 8 = y^2 - 2x$$

$$\Rightarrow x^2 + 2x - 8 = 0$$

$$\Rightarrow (x+4)(x-2)=0$$

$$\Rightarrow x = -4 \text{ or } 2.$$

But since, the point of intersection lies on the first quadrant, $x=2$.

By putting the value of x in any of the two curves, we can find the value of y.

$$\therefore y^2 = 2x$$

$$\Rightarrow y^2 = 4$$

$\Rightarrow y = 2$ or -2 . But the point of intersection is in first quadrant, so $y = 2$.

So, $P = (2, 2)$

Now, for the parabola $y^2 = 2x$, tangent can be found by differentiating the curve at $(2, 2)$.

$$\Rightarrow 2y \frac{dy}{dx} = 2$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2}$$

$\therefore$ Angle made by the tangent to parabola at P , $\tan \alpha = \frac{1}{2}$.

Similarly, tangent at the circle $x^2 + y^2 = 8$ can be found by differentiating the curve at $(2, 2)$

$$\Rightarrow 2x + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow 2y \frac{dy}{dx} = -2x$$

$$\Rightarrow \frac{dy}{dx} = \frac{-x}{y}$$

$$\Rightarrow \frac{dy}{dx} = -1.$$

$\therefore$ Angle made by the tangent to circle at P , $\tan \beta = -1$

Now, the acute angle between the two tangents is represented by $\alpha - \beta$.

We will use the expression $\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$

$$\Rightarrow \tan(\alpha - \beta) = \frac{\frac{1}{2} - (-1)}{1 - \frac{1}{2}}$$

$$\Rightarrow \tan(\alpha - \beta) = \frac{\frac{3}{2}}{\frac{1}{2}}$$

$$\Rightarrow \tan(\alpha - \beta) = 3$$

$\therefore \alpha - \beta = \tan^{-1}(3)$. therefore, option A

29. In an isosceles right triangle PQR , $\angle PRQ = 90^\circ$. The points S and T are two trisection points of QR . The value of $\tan(\angle SPT)$ is

A $\frac{3}{11}$

B $\frac{1}{11}$

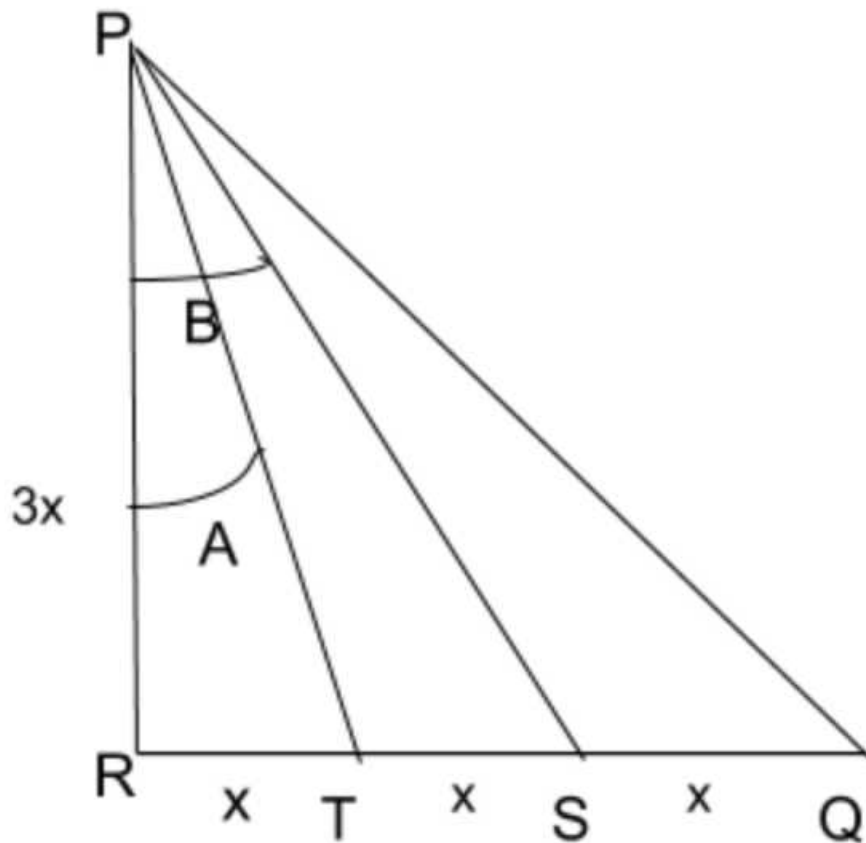
C $\frac{2}{3}$

D $\frac{1}{3}$

Answer: A

Explanation:

Let the isosceles sides be of the length $3x$ units each. Then, we can draw the triangle with given point as:



Angle $\angle SPT = B - A$.

$$\tan A = \frac{RT}{PR} = \frac{x}{3x} = \frac{1}{3}$$

$$\tan B = \frac{RS}{PR} = \frac{2x}{3x} = \frac{2}{3}$$

$$\text{So, } \tan(B - A) = \frac{\tan B - \tan A}{1 + \tan A \tan B}.$$

$$\Rightarrow \tan(B - A) = \frac{\frac{2}{3} - \frac{1}{3}}{1 + \frac{2}{9}}$$

$$\Rightarrow \tan(B - A) = \frac{\frac{1}{3}}{\frac{11}{9}} = \frac{3}{11}$$

30. Let $f : (0, \infty) \rightarrow (0, \infty)$ be a strictly decreasing function. Consider $h(x) = \frac{f\left(\frac{x}{1+x}\right)}{1+f\left(\frac{x}{1+x}\right)}$ Which one of the following is always true?

- A h is strictly decreasing
- B h is strictly increasing
- C h is strictly decreasing at first and then strictly increasing
- D h is strictly increasing at first and then strictly decreasing

Answer: A

Explanation:

$$h(x) = \frac{f\left(\frac{x}{1+x}\right)}{1+f\left(\frac{x}{1+x}\right)} = \frac{1+f\left(\frac{x}{1+x}\right)-1}{1+f\left(\frac{x}{1+x}\right)} = 1 - \frac{1}{1+f\left(\frac{x}{1+x}\right)} = 1 - \frac{1}{1+f\left(\frac{1}{1+x-1}\right)} = 1 - \frac{1}{1+f\left(1-\frac{1}{1+x}\right)}.$$

Now, Since f is a strictly decreasing function and its domain is positive, then, $f\left(\frac{1}{x+1}\right)$ will be a strictly increasing function. Again, $f\left(1-\frac{1}{x+1}\right)$ will be a strictly decreasing function.

Let us assume $1 + f\left(1-\frac{1}{x+1}\right) = g(x)$.

So, $g(x)$ is a strictly decreasing function.

$$h(x) = 1 - \frac{1}{g(x)}.$$

$\frac{1}{g(x)}$ becomes strictly increasing function, but $1 - \frac{1}{g(x)}$ will finally be strictly decreasing function.

Therefore, $h(x)$ becomes strictly decreasing function- Option A.

31. How many distinct 5×5 matrices are there such that each entry is either 0 or 1 and each row sum and each column sum is 4?

- A** 64
- B** 32
- C** 120
- D** 96

Answer: C

Explanation:

With either 0 or 1 as an entry, a sum of 4 is possible only with four 1s and one Zero.

So, each column and row must have four 1s and one 0.

Let us consider the first row, we can put the 0 in any of the 5 places.

So, number of ways of placing 1 zero in the first row=5.

In the second row, we cannot put the zero in the same column as that put by the first, as it will result in two 0s in the same column. So, the second 0 can be put in any of the remaining 4 places.

Similarly, for the third zero, we will have 3 ways to do so. And so on.

So, total number of ways to put 5 zeroes in the 5×5 matrix is $5 \times 4 \times 3 \times 2 \times 1 = 120$.

32. The sum of an infinite geometric series of real numbers is 14, and the sum of the cubes of the terms of this series is 392. The first term of the series is

- A** -14
- B** 10
- C** 7
- D** -5

Answer: C

Explanation:

Let the first term of the infinite G.P. be a and the common ratio be r .

So, Sum of infinite GP = $\frac{a}{1-r} = 14$ (1)

The cubes of the terms of original GP = $a^3, (ar)^3, (ar^2)^3 \dots = a^3, a^3r^3, a^3r^6 \dots$

So, in the new infinite GP, first term is a^3 and the common ratio is r^3

Therefore, sum to infinite terms = $\frac{a^3}{1-r^3}$ (2)

Cubing the eqn 1 and dividing by eqn 2, we get,

$$\frac{a^3}{(1-r)^3} = \frac{14 \times 14 \times 14}{1-r^3}$$

$$\Rightarrow \frac{1-r^3}{(1-r)^3} = \frac{14 \times 14 \times 14}{392}$$

$$\Rightarrow \frac{(1-r)(1+r^2+r)}{(1-r)^3} = 7$$

$$\Rightarrow \frac{(1+r^2+r)}{(1-r)^2} = 7$$

$$\Rightarrow (1+r^2+r) = 7(1+r^2-2r)$$

$$\Rightarrow 1+r^2+r = 7+7r^2-14r$$

$$\Rightarrow -6-6r^2+15r = 0$$

$$\Rightarrow 2r^2-5r+2 = 0$$

$$\Rightarrow 2r^2-4r-r+2 = 0$$

$$\Rightarrow 2r(r-2)-1(r-2) = 0$$

$$\Rightarrow (2r-1)(r-2) = 0$$

$$\therefore r = 1/2 \text{ or } 2.$$

In case of infinite GP, $-1 < r < 1$.

$$\text{So, } r = 1/2$$

Putting this value in eqn 1, we get,

$$\frac{a}{1-\frac{1}{2}} = 14$$

$$\Rightarrow 2a = 14$$

$$\therefore a = 7 \text{ - Option C.}$$

33. The radius of the incircle of the triangle formed by the x-axis and the lines $3x + 4y - 24 = 0$, $3x - 4y + 24 = 0$ is

A $\frac{7}{3}$

B $\frac{8}{3}$

C 2

D 3

Answer: B

Explanation:

For $3x+4y-24=0$ can be re-written as $3x + 4y = 24$

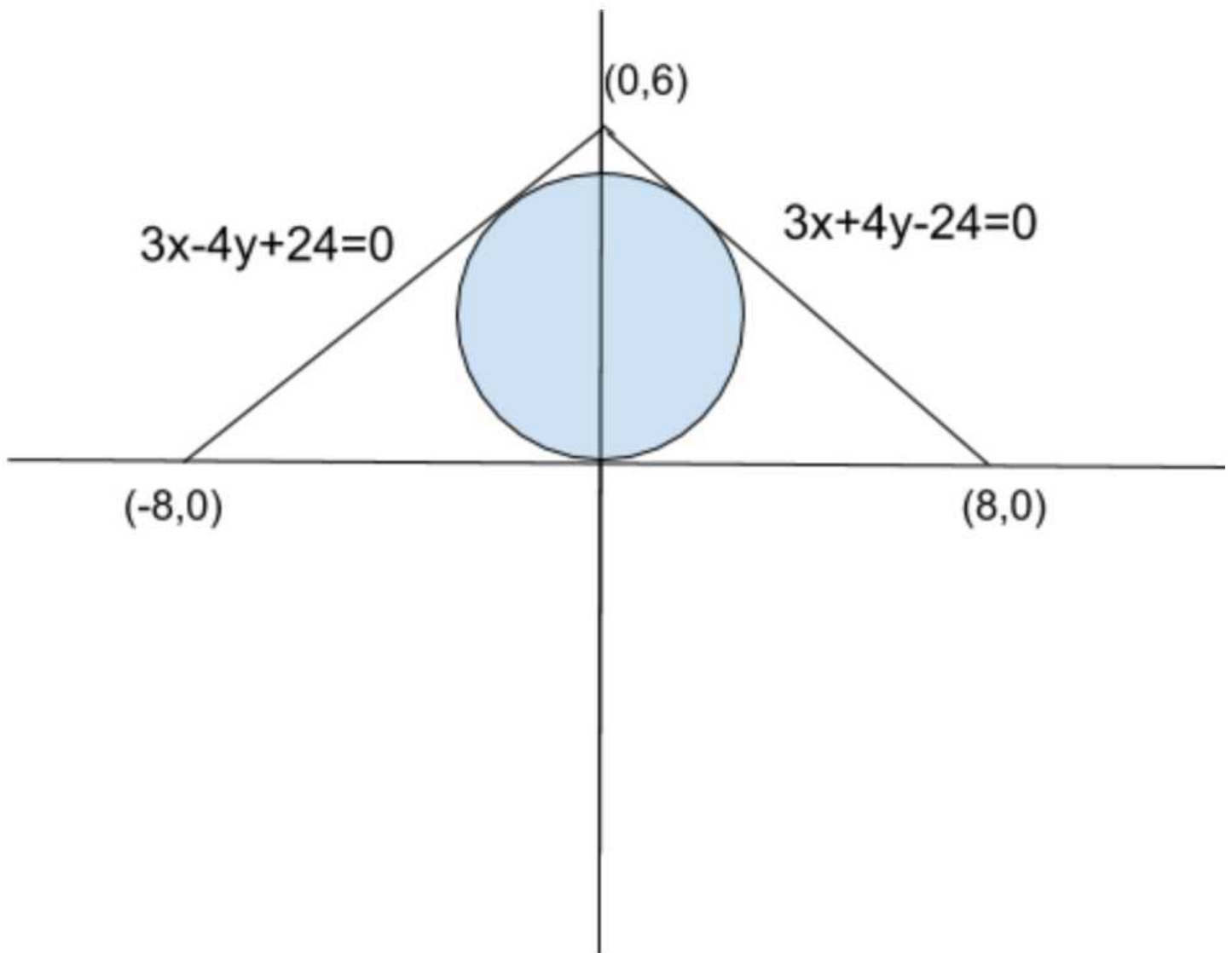
$$\Rightarrow \frac{3x+4y}{24} = 1$$

$$\Rightarrow \frac{x}{8} + \frac{y}{6} = 1.$$

So, this becomes the equation of the straight line of the form $\frac{x}{a} + \frac{y}{b} = 1$, where a and b are x and y intercepts respectively. The X and Y intercepts for this equation are 8 and 6 respectively.

Similarly $3x-4y+24=0$ can be re-written as $\frac{x}{-8} + \frac{y}{-6} = 1$. The X and Y intercepts for this equation are 8 and -6 respectively.

So, we get the following triangle and the incircle can be drawn as:



So, the triangle has sides of length 10,10 and 16.

Area of the triangle = $\frac{1}{2} \times \text{Base} \times \text{Height} = \frac{1}{2} \times 16 \times 6 = 48$

We know that $r \cdot s = \text{Area of the triangle}$, where r is the inradius and s is the semi-perimeter. $s = \frac{1}{2}(16 + 10 + 10) = 18$

So, $18r = 48$

$\Rightarrow$ Or, $r = \frac{48}{18} = \frac{8}{3}$

34. The expression $\tan^{-1}\left(\frac{1}{1+1.2}\right) + \tan^{-1}\left(\frac{1}{1+2.3}\right) + \tan^{-1}\left(\frac{1}{1+3.4}\right) + \dots + \tan^{-1}\left(\frac{1}{1+n(n+1)}\right)$ simplifies to

A $\tan^{-1}(1) - \tan^{-1}(n+1)$

B $\tan^{-1}(1) - \tan^{-1}(n)$

C $\tan^{-1}(n) - \tan^{-1}(1)$

D $\tan^{-1}(n+1) - \tan^{-1}(1)$

Answer: D

Explanation:

We will use the formula $\tan^{-1}\alpha - \tan^{-1}\beta = \tan^{-1}\left(\frac{\alpha - \beta}{1 + \alpha\beta}\right)$

The given equation can be re-written as $\tan^{-1}\left(\frac{2-1}{1+1.2}\right) + \tan^{-1}\left(\frac{3-2}{1+2.3}\right) + \dots + \tan^{-1}\left(\frac{(n+1)-n}{1+n(n+1)}\right)$
 $= (\tan^{-1}2 - \tan^{-1}1) + (\tan^{-1}3 - \tan^{-1}2) + \dots + (\tan^{-1}(n+1) - \tan^{-1}n)$
 $= \tan^{-1}(n+1) - \tan^{-1}1$

35. Let $g(x) = f(x) + f(2+x)$, where $f(x) = \begin{cases} 1 - |x|, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$ The number of points where the function g is not differentiable is

A 4

B 5

C 6

D 7

Answer: B

Explanation:

We will break the function into different parts and then plot the graph.

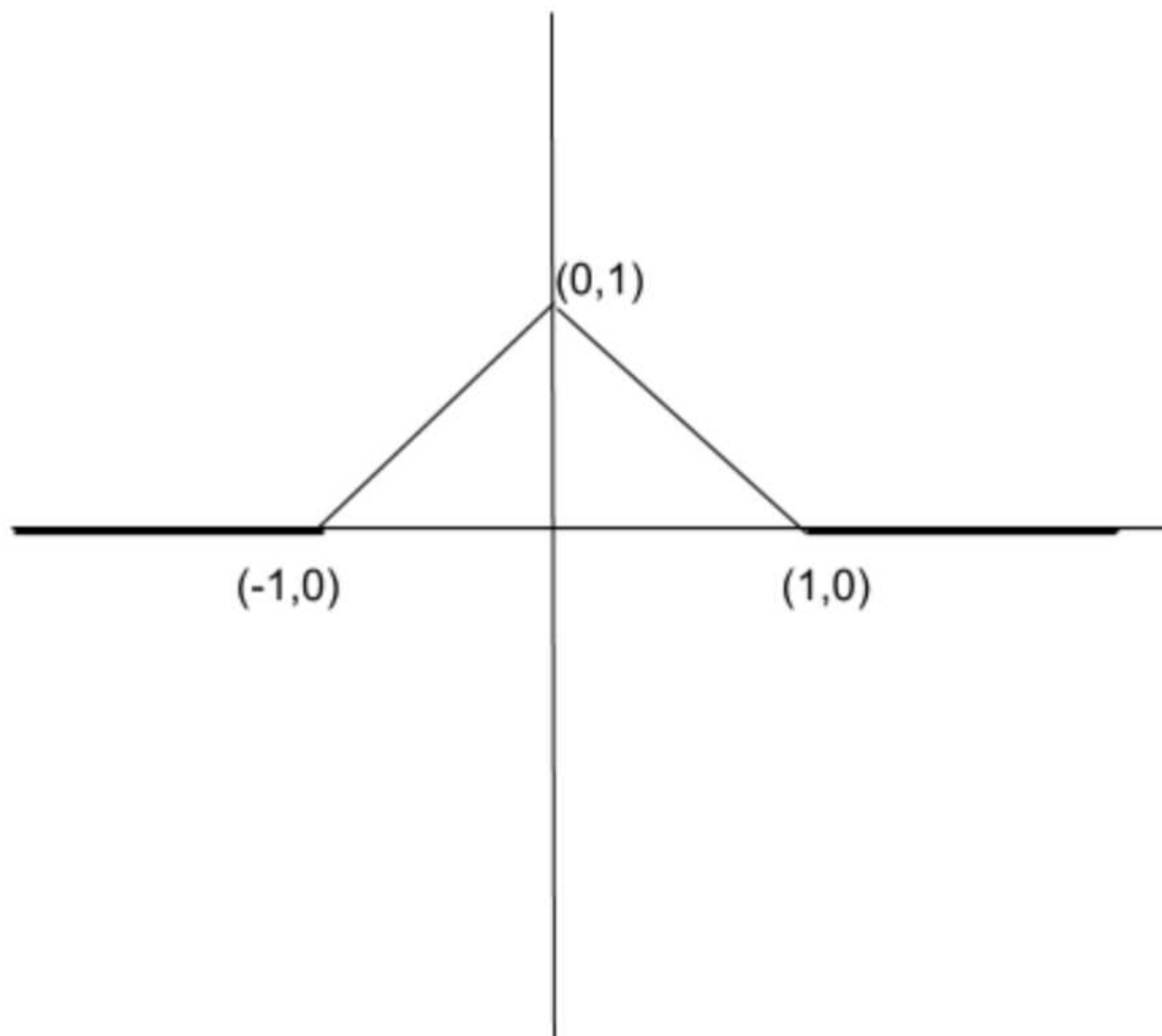
$f(x) = 0; x \in (-\infty, -1)$

$f(x) = 1 + x; x \in [-1, 0)$

$$f(x) = 1 - x; x \in [0, 1]$$

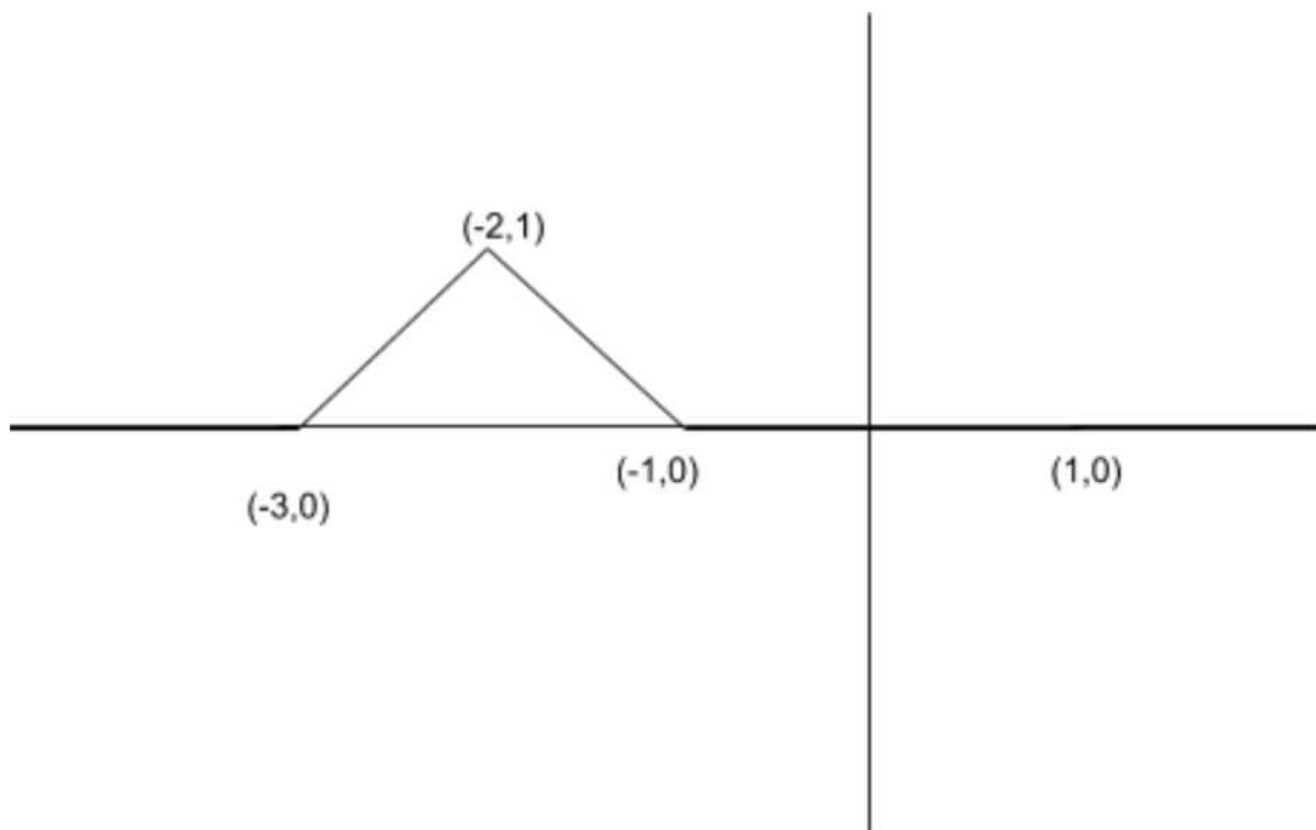
$$f(x) = 1 - x; x \in (1, \infty)$$

$f(x) =$



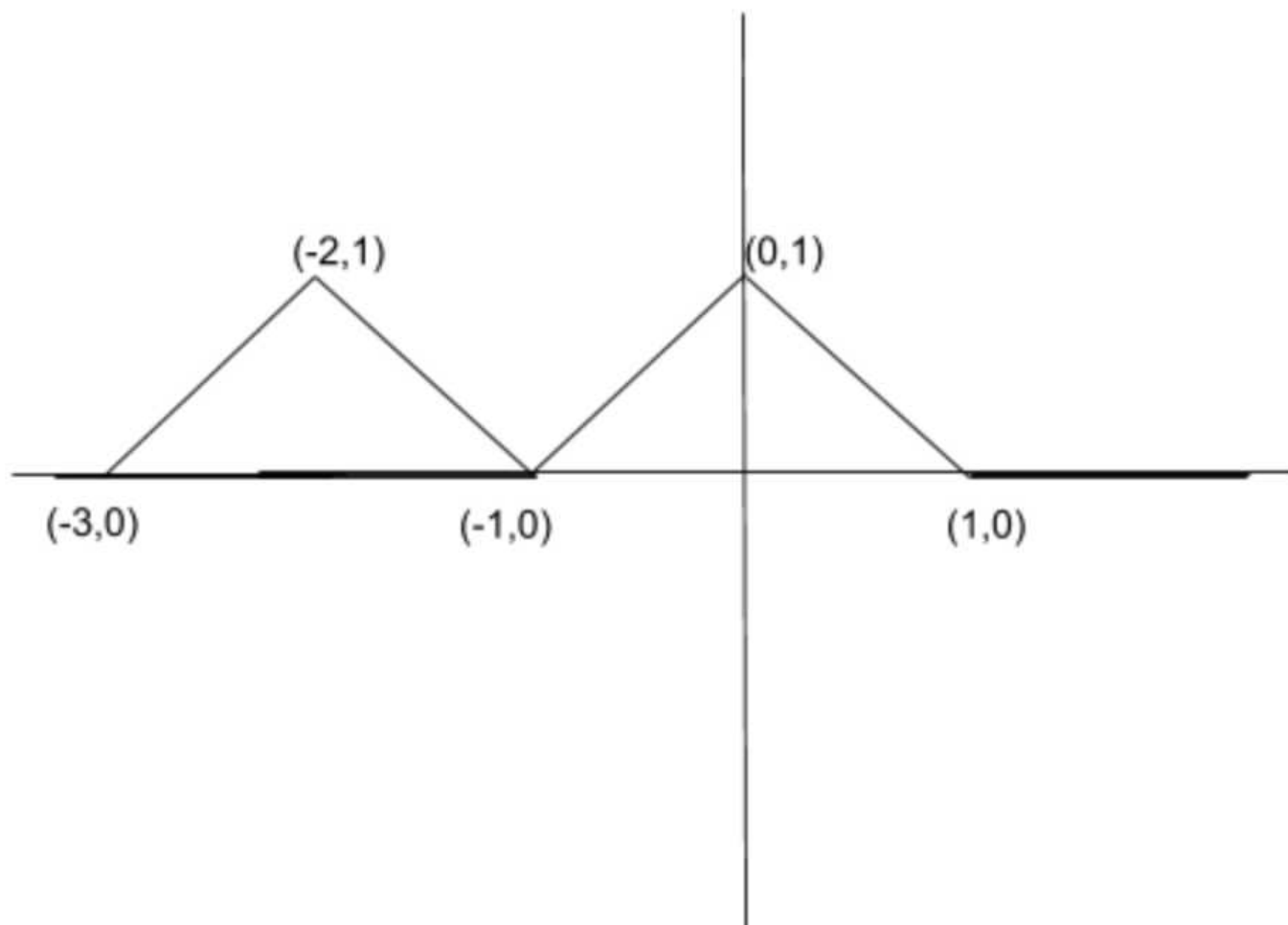
$f(x+2)$ is same as $f(x)$, but shifted 2 units to the left.

So, $f(x+2)$ can be drawn as:



We can observe that the parts where $f(x+2)$ takes non zero value only where $f(x)=0$. So, The graph of $g(x) = f(x) + f(x+2)$ will be superimposed upon each other, and there won't be any change in values.

The graph of $g(x)$ will be:



Now, $g(x)$ will be non-differentiable at sharp corners, because at those points Left Hand limit will not equal

to Right hand limit.

The graph of $g(x)$ has 5 such sharp edges at $(-3,0)$, $(-2,1)$, $(-1,0)$, $(0,1)$ and $(1,0)$. Therefore $g(x)$ is non-differentiable at 5 points.

36. If $A = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$, then A^{50} is

A $\begin{pmatrix} 50 & 0 \\ -1 & 1 \end{pmatrix}$

B $\begin{pmatrix} 1 & 0 \\ -51 & 1 \end{pmatrix}$

C $\begin{pmatrix} 1 & 0 \\ -50 & 1 \end{pmatrix}$

D $\begin{pmatrix} 1 & 0 \\ -49 & 1 \end{pmatrix}$

Answer: C

Explanation:

Given $A = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$, then A^{50}

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} (1)(1) + 0(-1) & 1(0) + 0(1) \\ (1)(-1) + (1)(-1) & (-1)(0) + (1)(1) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}. \end{aligned}$$

$$\begin{aligned} \text{Now, } A^3 &= \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} (1)(1) + 0(-1) & 1(0) + 0(1) \\ (1)(-2) + (1)(-1) & (-2)(0) + (1)(1) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}. \end{aligned}$$

So, in general $A^n = \begin{bmatrix} 1 & 0 \\ -n & 1 \end{bmatrix}$.

Therefore, $A^{50} = \begin{bmatrix} 1 & 0 \\ -50 & 1 \end{bmatrix}$, which is given in Option C

37. A curve is drawn such that the slope at any point $P = (x, y)$ is equal to x . The curve represents a family of

- A Circles
- B Parabolas
- C Ellipses
- D Hyperbolas

Answer: B

Explanation:

It is given that the slope of the curve at any point P is equal to x .

It is also known that, to find the slope at any point $P(x, y)$, we can find $\frac{dy}{dx}$ of the curve at that point P .

For $\frac{dy}{dx}$ of a curve to be equal to x , the original curve must be of the form $ax^2 + c$, where a and c are any constants.

Now, $y = ax^2 + c$ represents a family of parabolas.

Hence, option B

38. Let f be a differentiable function on $[-2, 2]$ such that $f(-2) = 1$, $f(2) = 5$ and $\left| \frac{df(x)}{dx} \right| \leq 1$ for all $x \in [-2, 2]$. The value of $f(0)$ is

- A 0
- B 1
- C 2
- D 3

Answer: D

Explanation:

Given that $\left| \frac{df(x)}{dx} \right| \leq 1$

Hence $-1 \leq \frac{df(x)}{dx} \leq 1$. Taking definite integral from $[-2, 2]$

$$-1 \cdot (5 - 1) \leq f(2) - f(-2) \leq (5 - 1)$$

$-4 \leq 4 \leq 4$. It is only possible when $\frac{df(x)}{dx}$ is always equal to 1. Therefore $f(x)$ is a straight line with

slope 1

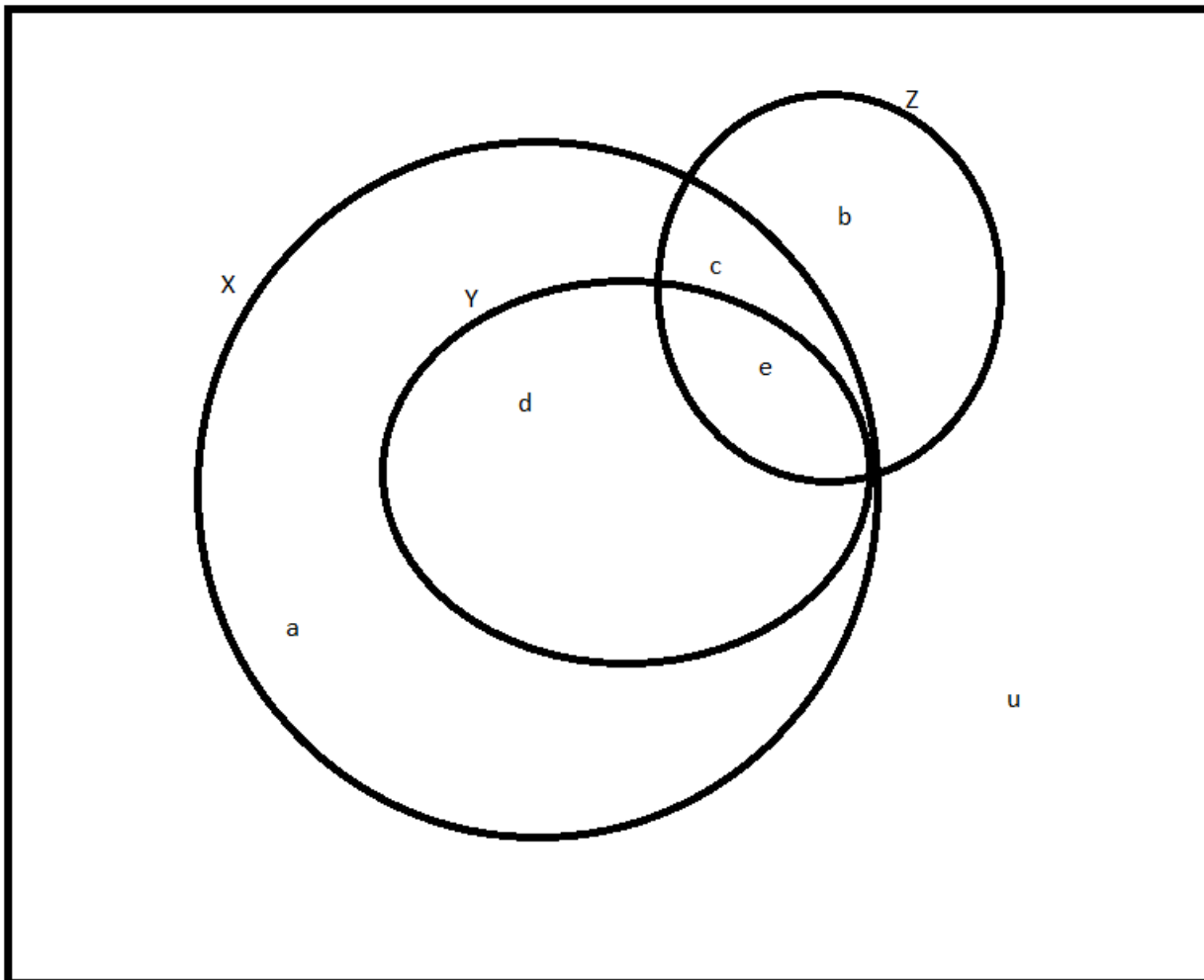
$$f(x) = x+3, f(0) = 3$$

39. For a set S , we denote by S' , the complement of the set S . Let X, Y, Z be Sets such that $Y \subseteq X$. Which of the following is always true?

- A $X \cap Z \subseteq Y \cap Z$
- B $Y' \cap Z' \supseteq X' \cap Z'$
- C $X \cap (Y \cup Z) = Y \cup (X \cap Z)$
- D $X' \cap Z \supseteq Y' \cap Z$

Answer: B

Explanation:



Consider the above Diagram

For option A $X \cap Z = c+e$ and $Y \cap Z = e$. Hence option A is False

For Option B $Y' \cap Z' = a+u$ and $X' \cap Z' = u$. Option B is True

For Option C $X \cap (Y \cup Z) = c+d+e$ and $Y \cup (X \cap Z) = c+d+e$. Option C is True

For option D $X' \cap Z = b$ and $Y' \cap Z = b+c$. Option D is False

40. A sequence $\{x_n\}$ of real numbers is defined as follows:

$$x_0 = 1, x_1 = 2, \text{ and } x_n = \frac{1+x_{n-1}}{x_{n-2}} \text{ for } n = 2, 3, 4 \dots$$

It follows that x_{2018} is

- A 1
- B 2
- C 3
- D 4

Answer: B

Explanation:

Given that $x_0 = 1$ and $x_1 = 2$

From above formula we get $x_2 = 3, x_3 = 2, x_4 = 1, x_5 = 1$ and $x_6 = 2$

We see that the pattern repeats itself in the period of five with $x_{5k} = 1, x_{5k+1} = 2, x_{5k+2} = 3, x_{5k+3} = 2$ and $x_{5k+4} = 1$

2018 is of form $5k+3$ hence $x_{2018} = 2$

41. The number of real roots of the equation

$$2 \sin \left(\frac{x^2+x}{6} \right) = 2^x + 2^{-x} \text{ is}$$

- A 0
- B 1
- C 2
- D ∞

Answer: A

Explanation:

In RHS of the equation,

The min value is attained is 2 when $x = 0$ (from A.M-GM inequality)

For the LHS side the max value is 2 when sin term has value of 1. But it can not be obtained when $x=0$

Hence, there are no real solution for the given equation

42. The number of points where the function

$$f(x) = \begin{cases} (x+1)^4 & x \leq 1 \\ (x-5)^2 & x > 1 \end{cases} \text{ attains its local maximum is}$$

- A 0

- B** 1
- C** 2
- D** 3

Answer: B

Explanation:

For $x \leq 1$ the graph of the function is an upward parabola that attains its min at $x = -1$ and value is 0, at $x = 1$ its value is 16

Similarly for $x > 1$ we see that its upward parabola which has value of 16 at $x = 1$ and decreases till $x = 5$ at which its value is 0. After which the values increases again.

It can be seen local maxima is attained at $x = 1$ only

43. A new sequence is obtained from the sequence of positive integers $\{1, 2, 3, \dots\}$ by deleting all the perfect squares. The 2018^{th} term of the new sequence is

- A** 2061
- B** 2062
- C** 2063
- D** 2065

Answer: C

Explanation:

Let us write all the number the numbers from 1 to 2018. Here we find that there are 44 numbers that are perfect squares which are $1, 4, 9, \dots, 1936(44^2)$. Thus after removing the perfect square it has total of $2018 - 44 = 1974$ terms.

Taking next 44 terms from 2019, it will be $(2019, 2020, 2021, \dots, 2062)$ in which there is 1 perfect square (2025 which is 45^2) hence it is removed and now there is total of 2017 terms. Next term has to be 2063 which is also 2018^{th} term

44. Consider the function

$$f(x) = \max\{e^x, e^{-x}\}, x \in \mathbb{R} \text{ It follows that}$$

- A** f is not continuous at some points
- B** f is continuous everywhere, but not differentiable anywhere
- C** f is continuous everywhere, but not differentiable at exactly one point
- D** f is differentiable everywhere

Answer: C

Explanation:

For $x < 0$, $f(x) = 2e^x$ and it is a continuous function with approaching the value 2 as $x \rightarrow 0$

For $x \geq 0$, $f(x) = 2e^{-x}$ and it is a continuous function with approaching the value 2 as $x \rightarrow 0$

Therefore the function is continuous.

However, $x \rightarrow 0^- f'(x) = 2$ but $x \rightarrow 0^+ f'(x) = -2$

Thus it is not differentiable at exactly 1 point

45. Let n be the number of ways in which 5 men and 6 women can stand in a queue such that all the women stand consecutively. Let m be the number of ways in which the same 11 persons can stand in a queue such that exactly 5 women stand consecutively. The value of $\frac{m}{n}$ is

A 5

B 6

C $\frac{5}{6}$

D $\frac{6}{5}$

Answer: A

Explanation:

For n , we consider B as a group that has all 6 women. Total arrangement of B and 5 men is $6!$. Further within B women can rearrange themselves in $6!$ ways.

Thus $n = 6! \times 6!$

For m , consider a group B_1 which has 5 women and B_2 has remaining 1 woman. Arrangement of B_1, B_2 and 5 men can be done in $7!$ ways. From this we need to subtract the permutation in which B_1 and B_2 are together. Which is $2! \times 6!$. Total ways = $7! - 2! \times 6!$. Which equals $5 \times 6!$. Furthermore in B_1 the selection and arrangement of 5 women can be done in ${}^6C_5 \times 5! = 6!$ ways.

$$m = 5 \times 6! \times 6!$$

$$\frac{m}{n} = 5$$

46. The locus of the centre of a circle that passes through the origin and cuts off a length $2a$ from the line $y = c$ is

A $x^2 + 2cx = a^2 + c^2$

B $x^2 + 2cy = a^2 + c^2$

C $y^2 + 2cx = a^2 + c^2$

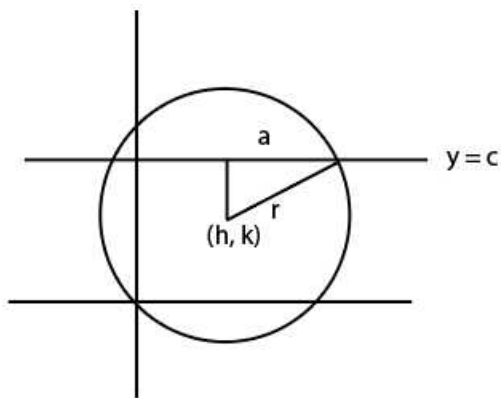
D $y^2 + 2cy = a^2 + c^2$

Answer: B

Explanation:

Let the locus of the center be represented by (h, k)

Since it passes through origin. Radius $r = \sqrt{h^2 + k^2}$



Since it cuts off chord of length = $2a$ on $y=c$. We draw 1 perpendicular bisector of the chord till the center and make a line which is intersecting point of line $y=c$ and the circle.

We obtain a right triangle that has " r " as its hypotenuse and a and $|c-k|$ as its sides.

Applying Pythagoras theorem,

$$r^2 = a^2 + (c - k)^2$$

it implies

$$h^2 + k^2 = a^2 + (c - k)^2$$

$$h^2 + k^2 = a^2 + c^2 + k^2 - 2ck$$

$$h^2 + 2ck = a^2 + c^2$$

replacing (h,k) by (x,y)

$$\text{we get } x^2 + 2cy = a^2 + c^2$$

47. Consider the function $f(x) = [x + 1] - \sin\left(\frac{\pi}{2}[x]\right)$ for $x \in \mathbb{R}$, where $[x]$ denotes the greatest integer less than or equal to x . Let $l_1 = \lim_{x \rightarrow 0^-} f(x)$ and $l_2 = \lim_{x \rightarrow 0^+} f(x)$ It follows that

- A $l_1 = l_2 = 1$
- B $l_1 = l_2 = -1$
- C $l_1 = -1; l_2 = 1$
- D $l_1 = 1; l_2 = -1$

Answer: A

Explanation:

$$l_1 = \lim_{x \rightarrow 0^-} f(x)$$

When $x \rightarrow 0^-$ then $[x] = -1$

$f(x)$ can be written as $[x] + 1 - \sin\left(\frac{\pi}{2}[x]\right)$.

Putting $[x] = -1$ we get $l_1 = -1 + 1 - \sin\left(\frac{\pi}{2} \times (-1)\right) = -\sin\left(\frac{\pi}{2} \times (-1)\right) = 1$

for l_2 , $x \rightarrow 0^+$ thus $[x] = 0$

putting it in $f(x)$. $l_2 = 0+1-0 = 1$

Hence $l_1 = l_2 = 1$

48. Consider the following system of equations:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ a & 9 & b & 10 \\ 6 & 8 & 10 & 13 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

The locus of all $(a, b) \in \mathbb{R}^2$ such that this system has at least two distinct solutions for (x_1, x_2, x_3, x_4) is

- A a parabola
- B a straight line
- C entire $\mathbb{R}^2$
- D a point

Answer: B

Explanation:

Let the above equation be written as $Ax=b$

where $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ a & 9 & b & 10 \\ 6 & 8 & 10 & 13 \end{bmatrix}$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$b = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

It is evident that $x_1 = x_2 = x_3 = x_4 = 0$ is a solution for the question.

It is stated that the equation has at least 2 distinct solution which implies infinite solution exists. and thus $\det(A) = 0$.

Calculating $\det(A)$ we get $\det(A) = 4(a+b-18)$

Equating it with 0, we get $a+b-18=0$ which is a straight line equation

49. The area bounded by the curve $y^2 = x^2 - x^4$ is

A $\frac{4}{3}$

B $\frac{2}{3}$

C $\frac{3}{4}$

D $\frac{3}{2}$

Answer: A

Explanation:

Since we see that both y and x has even terms, it implies the curve is symmetric along x-axis as well as y-axis.

Required area = 4*(area in quadrant I)

Area in quadrant I is given by

$$A = \int_0^1 \sqrt{x^2 - x^4} dx$$

$$A = \int_0^1 x \sqrt{1 - x^2} dx$$

take $1 - x^2 = t^2$, then $-2x dx = 2t dt$. when $x=0$ then $t=1$ and when $x=1$ then $t=0$

Updated integral can be written as

$$\int_0^1 t^2 dt = \left[\frac{t^3}{3} \right]_0^1 = \frac{1}{3}$$

$$\text{Required area} = 4*A = \frac{4}{3}$$

50. The sum of the infinite series

$$\cot^{-1} 2 + \cot^{-1} 8 + \cot^{-1} 18 + \cot^{-1} 32 + \cot^{-1} 50 + \dots \text{ is}$$

A $\frac{\pi}{3}$

B $\frac{\pi}{4}$

C $\frac{\pi}{6}$

D $\frac{\pi}{8}$

Answer: B

Explanation:

We see that 2,8,18... can be represented as $2n^2$ where $n \in \mathbb{N}$

$$\text{Required sum} = \sum_{n=1}^{\infty} (\cot^{-1} (2n^2)) = \sum_{n=1}^{\infty} (\tan^{-1} (\frac{1}{2n^2}))$$

$$= \sum_{n=1}^{\infty} (\tan^{-1} (\frac{2}{4n^2}))$$

$$\begin{aligned}
&= \sum_{n=1}^{\infty} \left(\tan^{-1} \left(\frac{(2n+1)-(2n-1)}{1+(4n^2-1)} \right) \right) \\
&= \sum_{n=1}^{\infty} \tan^{-1}(2n+1) - \tan^{-1}(2n-1) \\
&= \tan^{-1} \infty - \tan^{-1} 1 \\
&= \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}
\end{aligned}$$

